

1. Which of the following correctly distinguishes a statistic from a parameter? (2 Marks)
 - A. A statistic describes a population, a parameter describes a sample
 - B. A statistic describes a sample, a parameter describes a population
 - C. Both describe a population
 - D. Both describe a sample
2. Which of the following sampling methods is likely to have sampling bias? (2 Marks)
 - A. Voluntary response sampling
 - B. Stratified sampling
 - C. Random sampling
 - D. Cluster sampling
3. Which of the following is a key characteristic of the normal (Gaussian) distribution? (2 Marks)
 - A. It is skewed to the right
 - B. It is discrete and asymmetric
 - C. It is continuous and symmetric
 - D. It has a fixed variance of 1
4. Which condition is NOT required for the Central Limit Theorem? (2 Marks)
 - A. Independent samples
 - B. Population must be normally distributed
 - C. Large sample size ($n > 30$)
 - D. Identically distributed samples
5. Which term refers to a range of values used to estimate a population parameter? (2 Marks)
 - A. Point estimate
 - B. Confidence interval
 - C. Significance level
 - D. Test statistic
6. For coin flips: 3 heads, 2 tails. Model: $P(H)$. What is the likelihood? (2 Marks)
 - A. $P(H)^2 \times (1 - P(H))^3$
 - B. $P(H)^3 \times (1 - P(H))^2$
 - C. $P(H) \times (1 - P(H))$
 - D. $(P(H) \times (1 - P(H)))^5$

7. The core difference between MLE and MAP is: (2 Marks)

- A. MLE uses prior; MAP does not
- B. MAP uses prior; MLE does not
- C. Both maximize posterior
- D. No difference

8. If the Pearson correlation coefficient $r = 0$, which of the following conclusions is correct? (2 Marks)

- A. There is no relationship between the two variables
- B. There is no linear relationship between the two variables
- C. There is a strong nonlinear relationship between the two variables
- D. The two variables are completely independent

9. Which of the following correlation coefficients indicates the strongest linear relationship? (2 Marks)

- A. $r = 0.3$
- B. $r = -0.8$
- C. $r = 0.5$
- D. $r = -0.2$

10. In linear regression, if the residual is positive, it means that the actual value is _____ the predicted value. (2 Marks)

- A. Equal to
- B. Greater than
- C. Less than
- D. Unrelated to

11. In $\hat{y} = b_0 + b_1x$, b_0 is the: (2 Marks)

- A. Slope
- B. Coefficient
- C. Intercept
- D. Mean

12. The most commonly used distance metric in kNN is: (2 Marks)

- A. Cosine distance
- B. Sine distance
- C. Euclidean distance

D. Correlation distance

13. The two core steps of k means are: (2 Marks)

A. Calculate entropy; split nodes

B. Assign samples; update centroids

C. Merge clusters; recalculate distance

D. Select k neighbors; vote

14. The formula $P(x|y) = \frac{\text{count}+1}{\text{total} + \text{number of categories}}$ represents Laplacian smoothing in Naive Bayes. What is its core function? (2 Marks)

A. Reduce model variance

B. Fix the zero-probability issue

C. Solve underfitting

D. Prevent overfitting

15. In SVM, the points with Lagrange multiplier $\alpha_i = 0$: (2 Marks)

A. are all support vectors

B. determine the decision boundary

C. have no effect on building the SVM model

D. lie exactly on the margin hyperplane

16. In hard-margin SVM, the objective function $\frac{1}{2} \|w\|^2$ is minimized to: (2 Marks)

A. reduce the number of features

B. maximize the margin between two classes

C. increase classification error

D. make all data points support vectors

17. The Z-score value depends on the raw score, population _____ and population standard deviation. (2 Marks)

A. Mean

B. standard deviation

C. mode

D. interquartile range

18. Which statement is incorrect? (2 Marks)

A. We use t-distribution when sample size $n < 30$

B. We use Z-distribution for large sample size

C. t-distribution is used when population standard deviation is unknown

D. Z-distribution is only used for small samples

19. In hypothesis testing, we Fail to reject H_0 if the p-value is _____ α (alpha). (2 Marks)

A. greater than

B. less than or equal to

C. not equal to

D. approximately equal to

20. When a decision tree reaches a leaf node that contains samples from multiple classes, the final prediction is usually determined by a _____ vote among the samples in that node. (2 Marks)

A. weighted

B. random

C. majority

D. unanimous

21. F_1 -score is the harmonic mean of which two classification metrics? (2 Marks)

A. Accuracy and Recall

B. Precision and Recall

C. Precision and Specificity

D. Accuracy and Precision

22. If a binary classifier makes predictions completely at random, its expected AUC score will be _____. (2 Marks)

A. 0

B. 1

C. 0.5

D. 0.25

23: Which of the following is a solution to the overfitting problem in machine learning? (2 Marks)

A. Laplacian correction

B. Increasing model complexity

C. Adding more training data

D. Using zero-probability correction

24. To protect data privacy while training AI models across multiple local devices, we can use _____ learning. (2 Marks)

- A. deep
- B. transfer
- C. federated
- D. reinforcement

25. Injecting maliciously crafted mislabeled samples into the training dataset to degrade model performance is called: (2 Marks)

- A. Data poisoning
- B. Perturbation attack
- C. Anonymisation
- D. Algorithm simplification

26. Emotion detection and speech recognition belong to _____. (2 Marks)

- A. Cognitive Intelligence
- B. Perception Intelligence
- C. Decision Intelligence
- D. Creative Intelligence

27. Speech recognition and image classification belong to which category of AI? (2 Marks)

- A. Computational Intelligence
- B. Perceptual Intelligence
- C. Cognitive Intelligence
- D. General Intelligence

28. Why does the Naive Bayes classifier make the "naive" assumption? (2 Marks)

- A. It ignores the prior
- B. It assumes features are conditionally independent given the class
- C. It only handles binary tasks
- D. It requires large datasets

29. A tutor wants to model the relationship between weekly study time x (hours) and exam score y (out of 100) using **simple linear regression**. (8 Marks)

Given the following data for 6 students:

Student	x	y
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1	2	55
2	3	60
3	4	62
4	5	68
5	6	72
6	8	78

1. Compute the least-squares estimates of the regression coefficients $\widehat{\beta}_1$ and $\widehat{\beta}_0$.
2. Write the fitted regression line and use it to predict y when x=7.

Regression formula (simple linear regression):

- Model: $\hat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x$
- Slope: $\widehat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{s_{xy}}{s_{xx}}$
- Intercept: $\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}$

30. Compare model selection criteria and cross validation when used to evaluate a Gaussian Mixture Model. What assumptions are used for model selection criteria and cross validation? (7 Marks)

31. A tutor recorded the **exam scores (out of 100)** for 20 students: (15 Marks in total)

- 12, 18, 23, 24, 27, 29, 31, 33, 35, 36, 41, 44, 45, 48, 52, 55, 58, 62, 70, 85

Tasks:

1. **Construct a histogram** to show: (6 marks)
 - mass distribution of the marks
 - cumulative distribution of the marks

where equal-width bins of 10 marks are used.

2. Using your five-number summary, **draw a box plot** and identify any **outliers** using the 1.5×IQR rule (show IQR, fences, and which values are outliers, if any). (5 Marks)
3. Based on the histogram and box plot together, comment on the distribution of score and propose a scheme of scaling that makes 20% of the students achieving 70 marks or beyond. (4 marks)

32. You are given the following dataset of 12 samples (x_1, x_2) with two features and a binary class label y: (15 Marks in total)

Sample Index	x_1	x_2	Class y	Sample Index	x_1	x_2	Class y
1	2	1	0	2	3	1	0
3	2	2	0	4	4	2	0
5	6	5	1	6	7	5	1
7	6	6	1	8	8	6	1
9	5	4	1	10	1	2	0
11	5	3	1	12	3	2	0

Use **3-fold cross-validation** to evaluate a **k-Nearest Neighbour (kNN)** classifier with **city-block distance**:

$$\mathcal{D}(P_i, P_j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}|$$

The folds are fixed as follows:

- **Fold 1 (test):** {1, 2, 3, 4}
- **Fold 2 (test):** {5, 6, 7, 8}
- **Fold 3 (test):** {9, 10, 11, 12}

Tasks:

1. For each fold, train on the other 8 samples and predict the class label for each test sample using 1-NN. Show the nearest neighbour (sample ID) and the computed distance for each prediction. (8 Marks)
2. For each fold, compute the fold accuracy of testing dataset. (4 Marks)
3. Compute the overall cross-validation accuracy (mean of fold accuracies). (3 Marks)